

LAB 1

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Introduction

This lab is about, how incident responder perform a live analysis of system using simple tools.

- **Background:**

An employee at Tata Consultancy Services downloaded a software utility program

LairNetPutty.exe, without taking consent from the company IT department. The IT department asked the incident response team to investigate this unknown software and also told them to determine the risk to the company. The below report shows the findings of the incident response team, as well as our recommendations for stopping the risk.

I want an introduction what the lab is about. Do not copy and paste the lab explanation please put it in your own words. (This is an explanation please delete this from your report)

Analysis

1)Static Analysis:

Prior to Executing the application:

- 1)What utility can be run on the Windows VM to monitor processes running on the system?
- 2)What utility can be run on the Windows VM to monitor network connections on the system?
- 3)What application could you run on Windows to capture network traffic from the system
- 4)What information can you gather about the application from reviewing it without any special tools?

Task Manager can be used to monitor the windows VM process. Before running the application, our team found that all the application are running well.

The below image shows us the monitoring of Windows machine using Task Manager.

Processes Performance App history Startup Users Details Services					
Name	Status	7% CPU	55% Memory	1% Disk	0% Network
Apps (4)					
> Google Chrome (12)		0%	231.4 MB	0 MB/s	0 Mbps
> Greenshot		2.4%	18.4 MB	0.1 MB/s	0 Mbps
> Notepad		0%	1.9 MB	0 MB/s	0 Mbps
▼ Task Manager		1.8%	16.3 MB	0.1 MB/s	0 Mbps
Task Manager					
Background processes (43)					
Application Frame Host		0%	6.7 MB	0 MB/s	0 Mbps
COM Surrogate		0%	1.4 MB	0 MB/s	0 Mbps
> COM Surrogate		0%	1.5 MB	0 MB/s	0 Mbps
> Cortana (3)		0%	121.9 MB	0 MB/s	0 Mbps
CTF Loader		0.4%	1.8 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	2.2 MB	0.1 MB/s	0 Mbps
Java Update Checker (32 bit)		0%	1.2 MB	0 MB/s	0 Mbps

We can see from above picture that all the applications are running fine without any problem.

NETSTAT can be used to monitor the network connection. Our team checked, all the connection that are established are to the Microsoft(.NET) or Windows owned machine.

The below fig shows the established network connection.

```

:\Users\student\Downloads\volatility_2.6_win64_standalone>NETSTAT
Active Connections

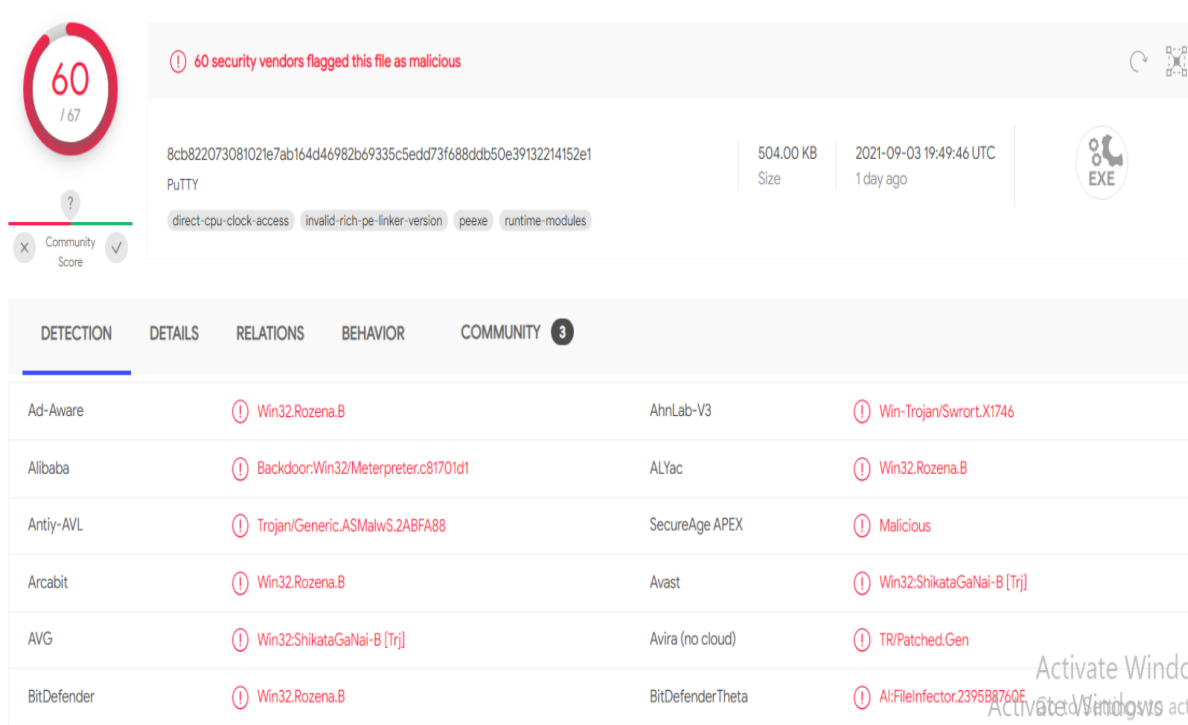
Proto Local Address           Foreign Address         State
TCP   192.168.74.158:55422     13.107.21.200:https     ESTABLISHED
TCP   192.168.74.158:55423     bingforbusiness:https  ESTABLISHED
TCP   192.168.74.158:55424     13.107.21.200:https     ESTABLISHED
TCP   192.168.74.158:55425     13.107.42.254:https     ESTABLISHED
TCP   192.168.74.158:55426     13.107.136.254:https    ESTABLISHED
TCP   192.168.74.158:55427     13.107.246.254:https    ESTABLISHED
TCP   192.168.74.158:55428     204.79.197.222:https    ESTABLISHED
TCP   192.168.74.158:61333     52.226.139.180:https    ESTABLISHED

:\Users\student\Downloads\volatility_2.6_win64_standalone>

```

Our team found that, Wireshark is one of the best applications to capture the network traffic.

After using the virus total to review the app, our team found that out of 67, 60 security vendors flagged this as Malicious. Next, we noticed that many of antivirus companies labeled this executable, The Trozen Malware. The Incident Response team done some internet searches to learn more about Trozen and found that it's a type of malware that is often disguised as legitimate software. Once activated in our machine, Trojan can enable cyber-criminals to spy on you, steal your sensitive data, and gain backdoor access to your system.



60 / 67

60 security vendors flagged this file as malicious

8cb822073081021e7ab164d46982b69335c5edd73f688ddb50e39132214152e1

PutTY

504.00 KB Size

2021-09-03 19:49:46 UTC 1 day ago

direct-cpu-clock-access invalid-rich-pe-linker-version peexe runtime-modules

DETECTION	DETAILS	RELATIONS	BEHAVIOR	COMMUNITY 3
Ad-Aware	Win32.Rozena.B	AhnLab-V3	Win-Trojan/Swrort.X1746	
Alibaba	Backdoor:Win32/Meterpreter.c81701d1	ALYac	Win32.Rozena.B	
Antiy-AVL	Trojan/Generic.ASMalwS.2ABFA88	SecureAge APEX	Malicious	
Arcabit	Win32.Rozena.B	Avast	Win32:ShikataGaNaI-B [Trj]	
AVG	Win32:ShikataGaNaI-B [Trj]	Avira (no cloud)	TR/Patched.Gen	
BitDefender	Win32.Rozena.B	BitDefenderTheta	AI:FileInfector.2395B8760E	

2) Dynamic Analysis

After launching the LairNetPutty.exe file, we got some suspicious errors.

While monitoring the app process using task manager, it seems everything works fine and below is the screenshot of the Task Manager.

Task Manager

File Options View

Processes Performance App history Startup Users Details Services

Name	Status	3% CPU	45% Memory	0% Disk	0% Network
Apps (5)					
> Notepad		0%	1.1 MB	0 MB/s	0 Mbps
> SSH, Telnet and Rlogin client (32...		0%	1.6 MB	0 MB/s	0 Mbps
> Task Manager		0%	15.9 MB	0 MB/s	0 Mbps
> Windows Command Processor (...)		0%	0.1 MB	0 MB/s	0 Mbps
> Wireshark		0%	11.9 MB	0 MB/s	0 Mbps
Background processes (33)					
Application Frame Host		0%	4.7 MB	0 MB/s	0 Mbps
COM Surrogate		0%	1.6 MB	0 MB/s	0 Mbps
> COM Surrogate		0%	0.9 MB	0 MB/s	0 Mbps
> Cortana (3)	⚙	0%	138.0 MB	0 MB/s	0 Mbps
CTF Loader		0%	1.7 MB	0 MB/s	0 Mbps
Greenshot		0%	2.8 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	1.0 MB	0 MB/s	0 Mbps

⬅ Fewer details End task

After checking the network connections using "NETSTAT" command, we found that some of the IP connections are getting connected to some malicious websites. We found two IPs are malicious by using internet searches and those IPs are 72.21.91.29, 72.21.81.200. These two IP's domains are Verizon.com.

```
C:\Users\student\Downloads\volatility_2.6_win64_standalone>NETSTAT
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	192.168.74.158:49596	iad23s94-in-f2:https	TIME_WAIT
TCP	192.168.74.158:52962	iad23s94-in-f4:https	TIME_WAIT
TCP	192.168.74.158:56491	52.226.139.185:https	ESTABLISHED
TCP	192.168.74.158:60668	172.67.203.114:http	TIME_WAIT
TCP	192.168.74.158:60669	172.67.203.114:https	TIME_WAIT
TCP	192.168.74.158:60672	13.89.179.9:https	TIME_WAIT
TCP	192.168.74.158:60673	13.107.21.200:https	ESTABLISHED
TCP	192.168.74.158:60674	bingforbusiness:https	ESTABLISHED
TCP	192.168.74.158:60675	13.107.21.200:https	ESTABLISHED
TCP	192.168.74.158:60676	13.107.4.254:https	ESTABLISHED
TCP	192.168.74.158:60677	72.21.81.200:https	ESTABLISHED
TCP	192.168.74.158:60678	204.79.197.254:https	ESTABLISHED
TCP	192.168.74.158:60679	204.79.197.222:https	ESTABLISHED
TCP	192.168.74.158:60681	131.253.33.254:https	ESTABLISHED
TCP	192.168.74.158:60682	13.107.136.254:https	ESTABLISHED
TCP	192.168.74.158:60683	a23-222-4-138:https	CLOSE_WAIT
TCP	192.168.74.158:61099	iad23s72-in-f2:https	TIME_WAIT

```
C:\Users\student\Downloads\volatility_2.6_win64_standalone>
```

Some Users comments about 72.21.91.29 are:

User Comments About 72.21.91.29

hepp help I need my tinfoil hat this IP is sending brainwave scans and controlling my mind through the net cables help me please its telling me to bury myself in my septic tank! - 2021-06-07

help the brainwave scans are getting worse from this ip its still telling me to bury myself in my septic tank i checmmk blacklist all green color of septic sewer waste making me crazy bad bad ip - 2021-06-07

На этот адрес ломился explorer.exe, похоже системный проводник был взломан - 2020-04-18

I keep getting cyberly stalked by this exact IP as well like all the others who reported about it on here. That same retard parasite is also using avast IP to hack my laptop as well that i found in my cmd. Is there anything i can do about it to stop it from happening. This has been going on for years now. - 2020-03-19

If you're using OpenSSL OpenVPN on OpenWRT, this may be the issue. There's a trojan OpenSSL1.1 around that acts normal but is far from it. - 2020-03-14

Uses multiple techniques to evade attacks and blocks once connected this address will literally take over most running processes and u will start seeing virtual machine addresses witch it will mask itself with ohh if u have this ip activity do not install any os images as they will be installed with back door system 32/agent p remote also Trojan droppers - 2020-01-02

discordhookhelper.exe connecting to 72/21/91.29 on http. nOpE. looks like spyware or a trojan. - 2019-08-09

discordhookhelper.exe connecting to 72.21.91.29 on http. nOpE. looks like spyware or a trojan. - 2019-08-09

2019.08.08 Bitdefender Firewall notification: The application C:\Users\...RarSFX7bddeploy.exe is sending a TCP(6) packet: ... to 72.21.91.29:HTTP(80) Due to the prior comments I will block. This is a Sony media center laptop and I have been unable to turn off all the MickeySoft spy log vacuuming, even though the available MS apps for that have been turned off. They don't turn them all off. Firewall blocking seems to be about the only way. IF that is what this is. - 2019-08-08

this ip needs to be investigated pronto !!! - 2019-06-20

About 72.21.81.200 are:

 Anonymous	08 Jul 2021	This ip continually tries to access my computer.	DNS Compromise DNS Poisoning Fraud Orders Phishing Fraud VoIP Open Proxy Email Spam Hacking Spoofing
 Anonymous	03 Jul 2021	webserver attack	Hacking Brute-Force Web App Attack
 Anonymous	02 Jul 2021	unexpected inbound activity on a likely compromised as set	Hacking
 Anonymous	23 Jun 2021	established connection	Hacking
 Anonymous	16 Jun 2021	Tried to connect to my computer. Ip showed during net s tat. Sent ping back it appears to be a server ... show more	Port Scan
 Anonymous	12 Jun 2021	When creating a microsoft account this ip connects to m y computer. It uses my person info to such as ... show more	DNS Compromise DNS Poisoning DDoS Attack FTP Brute-Force Hacking Spoofing Brute-Force
 Anonymous	23 May 2021	Hacking	DNS Compromise DNS Poisoning Fraud Orders Phishing Hacking

From the above pictures, we can see some users mention that, this IP is trying to get the system hacked and few mentioned it as a Trojan malware and trying to get the system access.

After capturing the network traffic using Wireshark. While running the LairNetputty.exe, the source is connecting to 192.168.74.158. Wireshark is marking the packets as a "Spurious Retransmission". So, our team find out more details about the Spurious retransmission.

No.	Time	Source	Destination	Protocol	Info
4	4.712731	Vmware_a1:d4:18	Vmware_ed:54:aa	ARP	Who has 192.168.74.2? Tell 192.168.74.158
5	4.713431	Vmware_ed:54:aa	Vmware_a1:d4:18	ARP	192.168.74.2 is at 00:50:56:ed:54:aa
6	4.728668	13.82.60.83	192.168.74.158	TCP	4444 → 57330 [RST, ACK] Seq=896078444 Ack=1 Win=64240 Len=0
7	5.228549	192.168.74.158	13.82.60.83	TCP	[TCP Spurious Retransmission] 57330 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
8	7.354279	13.82.60.83	192.168.74.158	TCP	4444 → 57330 [RST, ACK] Seq=1373044539 Ack=1 Win=64240 Len=0
9	7.355154	192.168.74.158	13.82.60.83	TCP	[TCP Port numbers reused] 57330 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
10	9.466978	13.82.60.83	192.168.74.158	TCP	4444 → 57330 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0
11	9.978252	192.168.74.158	13.82.60.83	TCP	[TCP Spurious Retransmission] 57330 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
12	12.076404	192.168.74.1	192.168.74.255	BROWSER	Host Announcement BHARATHREDDY, Workstation, Server, NT Workstation
13	12.084103	13.82.60.83	192.168.74.158	TCP	4444 → 57330 [RST, ACK] Seq=3989306694 Ack=1 Win=64240 Len=0
14	12.525215	192.168.74.158	192.168.74.2	NBNS	Refresh NB WINDOWS10<00>
15	12.587874	192.168.74.158	13.82.60.83	TCP	[TCP Spurious Retransmission] 57330 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
16	14.025307	192.168.74.158	192.168.74.2	NBNS	Refresh NB WINDOWS10<00>

> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
 > Ethernet II, Src: Vmware_a1:d4:18 (00:0c:29:a1:d4:18), Dst: Vmware_ed:54:aa (00:50:56:ed:54:aa)
 > Internet Protocol Version 4, Src: 192.168.74.158, Dst: 13.82.60.83
 > Transmission Control Protocol, Src Port: 57330 (57330), Dst Port: 4444 (4444), Seq: 0, Len: 0

```

0000  00 50 56 ed 54 aa 00 0c 29 a1 d4 18 08 00 45 00  .PV.T... )....E.
0010  00 34 25 43 40 00 08 06 00 00 c0 a8 4a 9e 0d 52  .4%Q@... ....J..R
0020  3c 53 df f2 11 5c 8a e2 e7 c3 00 00 00 00 00 02  <S...\.. ....
0030  fa f0 55 12 00 00 02 04 05 b4 01 03 03 08 01 01  ..U.....
0040  04 02  ..
  
```

Spurious Retransmission are the retransmission in which the receiver acknowledges the packet but not before the sender retransmits the packet (If you're seeing spurious retransmissions, it means that the sender thought packets were lost and sent them again, even though the receiver sent an acknowledge packet for it.)

Let's take a look inside the Wireshark's TCS dissector to see what the Wireshark development team wrote about Spurious Retransmissions.

```

2170      /* Check for spurious retransmission. If the current seq + segment length
2171      * is less than or equal to the current lastack, the packet contains
2172      * duplicate data and may be considered spurious.
2173      */
2174      if ( seglen > 0
2175      && tcpd->rev->tcp_analyze_seq_info->lastack
2176      && LE_SEQ(seq + seglen, tcpd->rev->tcp_analyze_seq_info->lastack) ) {
2177          if(!tcpd->ta){
2178              tcp_analyze_get_acked_struct(pinfo->num, seq, ack, TRUE, tcpd);
2179          }
2180          tcpd->ta->flags|=TCP_A_SPURIOUS_RETRANSMISSION;
2181          goto finished_checking_retransmission_type;
2182      }

```

The above picture clarifies that a spurious retransmission contains a Sequence number + Length Value that is less than the most recently received ACK(Acknowledgement).

Sometimes due to Retransmission Timeout, the ACK packet will be sent again to the host. But if it is repeating then Wireshark will mark it has a spurious retransmission. A few TCP spurious retransmission here and there are typically not a problem but, if it's starts increasing then we need to find the root cause for this problem.

If you are experiencing packet loss - look for an infrastructure device as the most likely cause of dropped ACKs. You will likely find that you also have "Previous Segment(s) Not Captured" throughout your trace file as that infrastructure device is probably dropping data packets as well as the ACK packets.

Conclusions

Firstly, I've learned about how to perform live analysis of a machine using simple tools and learned how to capture the network traffic using Wireshark and also how TCP and retransmission works. I've used NETSTAT to check the active connections and learned more parameters in NETSTAT.

Secondly, From the Discussion 1, I've learned about the VirusTotal and used that concept to review the LiarNetPutty.exe in this assignment. At first, I thought that there will be no errors because, I used putty and thought that this is another version in that application. As, I researched more about LiarNetPutty.exe and found that it's a malicious application.

Lastly, it's a great way to start the digital Forensic learning.

References

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